

Assignment: MCQ Questions based on Cars24 Case Study

1. Q1. What does `df.head(5)` return?

- A. First 5 rows
- B. Last 5 rows
- C. First 5 columns
- D. Entire dataset

2. Q2. Which method gives concise summary including column types and non-null counts?

- A. `df.describe()`
- B. `df.shape`
- C. `df.info()`
- D. `df.columns`

3. Q3. To view summary statistics for categorical columns, which is correct?

- A. `df.describe(include='object')`
- B. `df.describe(exclude='object')`
- C. `df.info()`
- D. `df.head()`

4. Q4. `df.tail(3)` shows:

- A. First 3 rows
- B. Last 3 rows
- C. Middle 3 rows
- D. Random 3 rows

5. Q5. Which of the following is NOT given by `df.describe()` for numerical data?

- A. Count
- B. Mean

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C. Median

D. Std

6. Q6. What does `df['mileage'].plot.density()` plot?

A. Histogram

B. Scatter

C. KDE plot

D. Bar chart

7. Q7. Which plot is best to visualize outliers?

A. Histogram

B. Line plot

C. Box plot

D. Scatter

8. Q8. What do the whiskers in a box plot represent?

A. Median and mean

B. Mean and std

C. Q1 and Q3

D. Minimum and maximum (excluding outliers)

9. Q9. Box plot with categorical x-axis and continuous y-axis shows:

A. Distribution of x

B. Relationship of categorical with numerical

C. Count of categories

D. Pairwise correlation

10. Q10. To visually inspect missing values pattern, best choice is:

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A. Heatmap

B. Histogram

C. KDE

D. Scatter

11. Q11. To replace all mileage values = 0 with NaN, use:

A. `df['mileage'] = None`

B. `df['mileage'].replace(0, np.nan)`

C. `df['mileage'] = 0`

D. `df.drop('mileage')`

12. Q12. To delete the 'full_name' column permanently:

A. `del df.full_name`

B. `df.drop('full_name')`

C. `df = df.drop('full_name', axis=1)`

D. `df.full_name.drop()`

13. Q13. Which metric is least affected by outliers?

A. Mean

B. Median

C. Mode

D. Standard Deviation

14. Q14. If 0.8% of rows have mileage = 0, what's the next step?

A. Drop entire dataset

B. Fill with 0

C. Consider replacing or dropping those rows

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D. Use them for predictions

15. Q15. Splitting full_name into make and model uses:

A. str.extract()

B. str.split()

C. apply(lambda

D. Both B and C

16. Q16. To create a column age from column year:

A. `df['age'] = 2020 - df['year']`

B. `df['age'] = df['year'] - 2020`

C. `df['age'] = df['year'] / 2020`

D. `df['age'] = 2020 + df['year']`

17. Q17. What's the result of `df.sample(frac=1)`?

A. Removes duplicates

B. Sorts data

C. Shuffles the dataset

D. Splits data

18. Q18. Which transformation reduces skewness in price data?

A. Square

B. Log

C. Mean normalization

D. Label encoding

19. Q19. Which is NOT an encoding method?

A. One-hot

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B. Label

C. Heatmap

D. Target

20. Q20. Which encoding causes high dimensionality?

A. Label

B. One-hot

C. Target

D. Binary

21. Q21. Which method assigns mean target value to category?

A. One-hot

B. Label

C. Target

D. Manual

22. Q22. What does `pd.get_dummies(df['fuel_type'], prefix='fuel')` do?

A. One-hot encode fuel_type

B. Label encode fuel_type

C. Drops fuel_type

D. Multiplies fuel_type

23. Q23. Which of the following uses a color map to show correlations?

A. Pairplot

B. Boxplot

C. Heatmap

D. Barplot

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24. Q24. To get correlation between price and age:

- A. `df['price'].corr(df['age'])`
- B. `corr(df['price'], df['age'])`
- C. `df.corr()['price']['age']`
- D. Both A and C

25. Q25. Which library is usually imported as `sns`?

- A. `matplotlib`
- B. `pandas`
- C. `seaborn`
- D. `numpy`

26. Q26. To rotate x-axis labels in seaborn barplot:

- A. `plt.xlim()`
- B. `plt.xticks(rotation=90)`
- C. `sns.rotate()`
- D. `plt.xlabel(rotate=90)`

27. Q27. What will this print?

```
model = full_name.apply(lambda x: ' '.join(x.split()[1:]))
```

- A. Brand name only
- B. Full name
- C. All except brand
- D. Error

28. Q28. What happens if the column doesn't exist: `df = df.drop('Automatic', axis=1)`?

- A. Error

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B. Silent pass

C. Creates column

D. Ignores

29. Q29. What's wrong in: `df = pd.read('data.csv')`?

A. read is not a function

B. Missing `read_csv()`

C. Should be `open()`

D. Nothing

30. Q30. Which is invalid for checking column names?

A. `df.columns`

B. `df.columns()`

C. `list(df.columns)`

D. `for col in df.columns:`

31. 31. What will `df['seats'].value_counts(normalize=True)` return?

A. Raw counts

B. % distribution

C. Median values

D. Mode values

32. 32. Which method helps visualize pairwise relationships with plots for multiple columns?

A. heatmap

B. pairplot

C. scatter

D. countplot

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33. 33. To combine two DataFrames horizontally, which method is used?

- A. append
- B. merge
- C. concat(axis=1)
- D. join

34. 34. What will `df.shape` return?

- A. Number of columns
- B. Size of dataset
- C. (rows, columns)
- D. Data types

35. 35. When should you use `.dropna()`?

- A. Always
- B. When too many nulls
- C. To drop duplicates
- D. To add new column

36. 36. What's wrong in: `df['price'].hist(bins=30).plot()`?

- A. Incorrect chaining
- B. Missing `plt.show()`
- C. Use seaborn instead
- D. Nothing

37. 37. What type of encoding is used here:

```
df['fuel_type'].map({'Petrol': 0, 'Diesel': 1})
```

- A. One-hot

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B. Target

C. Label

D. Binary

38. 38. Which code removes 'Manual' from a list of dummy columns correctly?

A. `df.drop('Manual')`

B. `df = df.drop('Manual', axis=1)`

C. `del df['Manual']`

D. All

39. 39. If a column has too many unique categories, which encoding is better?

A. One-hot

B. Label

C. Target

D. Random

40. 40. Which plot best compares prices across brands?

A. lineplot

B. heatmap

C. boxplot

D. scatter

41. 41. Which of these will cause error?

``df['col'].hist().density()``

A. `.hist` is not chainable

B. `.density` not a method

C. `.col` is invalid

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D. All

42. 42. To encode 'make' based on average price:

Which method is correct?

A. LabelEncoder

B. `df['make'].map(df.groupby('make')['selling_price'].mean())`

C. `pd.get_dummies()`

D. `np.mean(df['make'])`

43. 43. Which of these is not a pandas operation?

A. `df.append()`

B. `df.sort()`

C. `df.aggregate()`

D. `df.plot()`

44. 44. What will happen if you use ``.iloc[-1]``?

A. First row

B. Last row

C. Last column

D. Error

45. 45. In seaborn, to plot multiple KDEs by category:

A. `sns.kdeplot(data, hue='col')`

B. `sns.histplot(col='target')`

C. `sns.boxplot()`

D. `sns.scatter()`

46. 46. Why drop 'Automatic' after dummy encoding 'transmission_type'?

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- A. Reduce redundancy
- B. Avoid multicollinearity
- C. Prevent overfitting
- D. All

47. 47. Code:

```
`df['new'] = df['selling_price'].apply(np.log)`
```

What does it do?

- A. Removes skew
- B. Inverts values
- C. Filters price
- D. Errors

48. 48. How to sort df by price descending?

- A. `df.sort_values('price', ascending=True)`
- B. `df.sort_values('price', ascending=False)`
- C. `df['price'].sort()`
- D. `df.order_by('price')`

49. 49. `df['fuel'].value_counts()` returns?

- A. Total nulls
- B. Frequency of unique values
- C. Descriptive stats
- D. Unique column names

50. 50. Code:

```
`plt.figure(figsize=(10,5)); sns.boxplot(x='make', y='price', data=df)`
```

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Used to?

- A. Group brands by price
- B. Group price by years
- C. Show bar chart
- D. Heatmap comparison